# Lab 04: Networking

**What is Protocol?**

A network protocol is a set of rules that is set up by people that determine how a particular data is transmitted between different devices in the same network. e.g.: HTTP, TCP, IP, FTP, SMTP, etc.

**IP Address :**

An IP (Internet Protocol) address is a unique number assigned to each device on a network, allowing them to communicate with each other. It's like a device's "address" on the internet or local network.

**IP Address Parts:**

192.168.1.10/24

What /24 means:

- First 3 numbers are the network (192.168.1)
- Last number is the host (10)
- Can have 256 addresses (0-255)
- Usable: 1-254 (0 is network, 255 is broadcast)

**IP Address Types:**

| Type | Range | Who Uses It | Example |
|---|---|---|---|
| **Private (Home/Office)** | 192.168.0.0 - 192.168.255.255 | Local networks only | 192.168.1.10 |
| | 10.0.0.0 - 10.255.255.255 | Large organizations | 10.0.1.50 |

| Type | Range | Who Uses It | Example |
|---|---|---|---|
| | 172.16.0.0 - 172.31.255.255 | Medium organizations | 172.16.0.100 |
| **Public** | Everything else | Internet-facing | 8.8.8.8 (Google DNS) |
| **Loopback** | 127.0.0.1 | This computer only | 127.0.0.1 (localhost) |
| **Link-Local** | 169.254.x.x | Auto-assigned when DHCP fails | 169.254.1.1 |
| **Broadcast** | 255.255.255.255 | Send to all devices | Used for discovery |

**Special IP Addresses You'll See:**

| IP Address | What It Means | When You See It |
|---|---|---|
| **0.0.0.0** | All interfaces / Not configured | Server listening on all IPs, or no IP assigned |
| **127.0.0.1** | Localhost (this computer) | Testing local services |
| **192.168.1.1** | Typical home router | Your default gateway |
| **8.8.8.8** | Google's public DNS | DNS server, connectivity testing |
| **1.1.1.1** | Cloudflare's public DNS | Alternative DNS server |
| **255.255.255.255** | Broadcast to all | Network discovery |

**Common Subnet Masks:**

| CIDR | Subnet Mask | How Many Hosts | Typical Use |
|---|---|---|---|
| /32 | 255.255.255.255 | 1 (just this IP) | Single host |
| /24 | 255.255.255.0 | 254 | Home/small office |
| /16 | 255.255.0.0 | 65,534 | Large organization |
| /8 | 255.0.0.0 | 16,777,214 | ISP/Enterprise |

**What is a Port?**

- Virtual "door" or "channel" for network services
- Allows multiple services on same IP address
- Numbers from 0 to 65535

**How Ports Work:**

Your Computer Remote Server

192.168.1.10 93.184.216.34

Request:

192.168.1.10:54321 -------> 93.184.216.34:80

(random port) (web server)

Response:

192.168.1.10:54321 <------- 93.184.216.34:80

## Common Ports Beginners Must Know

| Service | Protocol | Port |
|---|---|---|
| SSH | TCP | 22 |
| HTTP | TCP | 80 |
| HTTPS | TCP | 443 |
| DNS | UDP,TCP | 53 |
| FTP | TCP | 20,21 |
| SMTP | TCP | 25 |
| MySQL | TCP | 3306 |
| PostgreSQL | TCP | 5432 |

**What is a Protocol?**

- Set of rules for network communication
- Like languages humans speak to understand each other
- Different protocols for different purposes

**Protocol Stack (How They Work Together):**

```
┌──────────────────────────────────────────┐
│ Application Layer          │ What: User services
│ HTTP, FTP, SSH, DNS, SMTP  │ Example: Web browsing
├──────────────────────────────────────────┤
│ Transport Layer            │ What: Port-to-port delivery
│ TCP, UDP                   │ Example: Reliable vs fast
```

```
├──────────────────────────────────────┤
│ Network Layer          │ What: IP addressing, routing
│ IP, ICMP               │ Example: Packet routing
├──────────────────────────────────────┤
│ Link Layer             │ What: Physical connection
│ Ethernet, Wi-Fi        │ Example: Local network
└──────────────────────────────────────┘
```

**Understanding DNS (Domain Name System)**

**What is DNS?**

- Translates human-readable names to IP addresses
- Like a phone book for the internet
- You type google.com, DNS returns 142.250.185.46

**How DNS Works (Step by Step):**

1. You type: google.com in browser

↓

2. Computer checks: /etc/hosts (local file)

↓

3. If not found, asks: DNS server (configured in /etc/resolv.conf)

↓

4. DNS server responds: 142.250.185.46

↓

5. Browser connects to: 142.250.185.46 on port 443

**Gateway/Router:**

- Connects your network to other networks (like the internet)
- Usually first IP in your subnet (192.168.1.1)
- All traffic to internet goes through it
- Think: The exit door of your building

**Network Interface:**

- Hardware/software connection point
- eth0 = first Ethernet
- wlan0 = wireless
- lo = loopback (127.0.0.1)
- Think: The network port on your computer

**MAC Address (Media Access Control):**

- Physical hardware address (48 bits)
- Format: 00:1A:2B:3C:4D:5E
- Unique to each network card
- Used at link layer (local network only)
- Think: Serial number of network card

**View Your Network Configuration**

**Command Reference: Network Info**

| Command | What It Shows | Example |
|---|---|---|
| ip addr or ip a | All network interfaces and IP addresses | ip a |
| ip link | Interface status (up/down) | ip link show |

| Command | What It Shows | Example |
| --- | --- | --- |
| ip route | Where packets go (routing table) | ip route show |
| hostname | Computer's name | hostname |
| hostname -I | All IP addresses of this machine | hostname -I |
| cat /etc/resolv.conf | DNS servers being used | cat /etc/resolv.conf |

Show all interfaces in brief format

```
ip -br addr show
```

Find your main IP address

```
hostname -I | awk '{print $1}'
```

Find your default gateway

```
ip route | grep default | awk '{print $3}'
```

Check DNS servers

```
grep nameserver /etc/resolv.conf
```

Check if interface is up

```
ip link show | grep "state UP"
```

Show only active interfaces

```
ip -br link show | grep UP
```

**Testing Network Connectivity**

**Command Reference: Connectivity Testing**

| Command | What It Does | When to Use |
|---|---|---|
| ping HOST | Test if host is reachable | First step in troubleshooting |
| ping -c 4 HOST | Send 4 packets only | Quick test |
| traceroute HOST | Show path packets take | Find where connection fails |
| tracepath HOST | Like traceroute (no root) | Alternative to traceroute |
| mtr HOST | Continuous traceroute | Monitor connection quality |

```
# Basic ping (runs forever, press Ctrl+C to stop)
ping google.com

# Send only 4 packets
ping -c 4 google.com

# Faster pings (every 0.2 seconds)
ping -c 10 -i 0.2 google.com

# Quiet mode (only show summary)
ping -c 10 -q google.com

# Test loopback (always works if network stack is OK)
ping 127.0.0.1
```

```
# Test your own IP
ping $(hostname -I | awk '{print $1}')

# Test your gateway
ping $(ip route | grep default | awk '{print $3}')

# Test internet (Google's DNS)
ping 8.8.8.8

# Test with name (checks DNS too)
ping google.com
```

```
# Basic traceroute
traceroute google.com

# Faster (no DNS lookups)
traceroute -n google.com

# Alternative (doesn't need root)
tracepath google.com

# Continuous traceroute (better for monitoring)
mtr google.com

# MTR with no DNS (faster)
mtr -n google.com
```

```
# MTR report mode (run 10 cycles and show summary)
mtr -r -c 10 google.com
```

**Checking Ports and Services**

**Command Reference: Port Checking**

| Command | What It Shows | Common Usage |
|---|---|---|
| ss -tuln | All TCP/UDP listening ports | ss -tuln |
| ss -tulnp | Include process info (needs root) | sudo ss -tulnp |
| ss -t | All TCP connections | ss -t |
| netstat -tuln | Legacy version of ss | netstat -tuln |
| lsof -i :PORT | What's using specific port | sudo lsof -i :80 |

```
# Modern command: ss (faster than netstat)
ss -tuln

# With process names (needs root)
sudo ss -tulnp

# Show only listening ports
ss -tuln | grep LISTEN

# Show only TCP
ss -tln
```

```
# Show only UDP
ss -uln

# Show all connections (not just listening)
ss -tun

# Find what's using a specific port
sudo ss -tulnp | grep :80

setting and listening
nc -l 8080
nc 127.0.0.1 8080
```

```
Using netcat (nc)
nc -zv localhost 80
nc -zv google.com 443
```

**DNS Troubleshooting**

**Command Reference: DNS Tools**

| Command | What It Does | When to Use |
|---|---|---|
| dig DOMAIN | Detailed DNS lookup | Troubleshooting, see all details |
| dig +short DOMAIN | Quick answer only | Just need the IP |
| nslookup DOMAIN | Simple DNS lookup | Quick checks |

| Command | What It Does | When to Use |
| --- | --- | --- |
| host DOMAIN | Basic DNS lookup | Simplest option |
| cat /etc/resolv.conf | Show DNS servers | Check what DNS you're using |

```
# Quick lookup - just show IP
dig +short google.com

# Detailed lookup
dig google.com

# Query specific DNS server
dig @8.8.8.8 google.com
dig @1.1.1.1 google.com

# Query specific record type
dig google.com A        # IPv4 address
dig google.com AAAA     # IPv6 address
dig google.com MX       # Mail servers

# Reverse lookup (IP to name)
dig -x 8.8.8.8



# Simple lookup with nslookup
nslookup google.com
nslookup google.com 8.8.8.8  # Use specific DNS server
```

**Firewalls & Security Groups**

**Commands & Use Cases**

| Command | Explanation | Scenario |
|---|---|---|
| sudo ufw status verbose | Check current rules | Daily check |
| sudo ufw allow 80/tcp | Allow HTTP from anywhere | Public web app |
| sudo ufw deny out 80/tcp | Block outgoing HTTP | Test firewall effect |
| sudo ufw reset | Clean everything | Lab cleanup |

Check Current Firewall Status:

sudo ufw status verbose

Enable the Firewall

sudo ufw enable

sudo ufw allow 80/tcp

sudo ufw deny 23/tcp

Check Firewall Rules

sudo ufw status numbered

sudo ufw delete 2

sudo ufw delete allow 443/tcp

Reset Firewall

sudo ufw reset

Disable Firewall

sudo ufw disable

**Exercise 1: View Network Configuration**

**Task:** Display your network interfaces, IP addresses, gateway, and DNS servers.

**Exercise 2: Test Connectivity to Remote Server**

**Task:** Test connectivity to google.com and trace the route.

**Types of IP Addresses**

1. **IPv4**:

- 32-bit address, written as four numbers separated by dots (e.g., 123.89.46.72).
- This is a 32-bit IP address, which means it contains a combo of 32 (1 and 0's). In this version of the IP address, there are 4 groups or **Octets** (8 bits), and each octet is represented by a decimal value in the address. It is easy to remember.
- Commonly used, but the limited number of addresses (about 4.3 billion).

**2. IPv6:**

- 128-bit address, written in eight groups of hexadecimal numbers (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- It provides a vastly larger pool of addresses, designed to replace IPv4 as it runs out.

**3. Public IP**

- Identifies devices on the internet.
- Internet Service Providers (ISPs) assign these. They are accessible worldwide.

**4. Private IP**
- Works within private networks like home or office networks.
- Not reachable from the internet. Usually appears as `192.168.x.x`, `10.x.x.x` or `172.16.x.x` — `172.31.x.x`.

**IP Address Classes (IPv4 Only)**

IPv4 addresses are separated into five classes. The starting number decides their use in networks.

Class A: Range 1.0.0.0 - 126.0.0.0.
Purpose: Very large networks, such as big organizations.

Class B: Range 128.0.0.0 - 191.255.0.0.
Purpose: Medium-sized networks.

Class C: Range 192.0.0.0 - 223.255.255.0.
Purpose: Small networks. These are often used in homes or business LANs.

Class D: Range 224.0.0.0 - 239.255.255.255.
Purpose: Reserved for multicasting.

Class E: Range 240.0.0.0 - 255.255.255.255.
Purpose: Experimental. Used for research.

**IP address — Network ID and Host ID**

An IP address has two parts: Network ID and Host ID. A Host refers to any device receiving an IP address.

1. ***The Network ID changes with the IP class:***
   i) Class A: The 1st octet is the Network ID.
   ii) Class B: The 1st and 2nd octets form the Network ID.
   iii) Class C: The 1st, 2nd, and 3rd octets create the Network ID.
2. Direct Connection: Devices with identical Network IDs connect without a router.
3. Router Need: Devices with different Network IDs require a router.

IP Address Structure and Device Connection

```
+--------------------+--------------------+--------------------+
| IP Address Class | Network ID       | Host ID          |
+--------------------+--------------------+--------------------+
| Class A          | 1st Octet        | 2nd, 3rd, 4th    |
```

```
|                    | Example: 17.0.0.1 |  (0.0.1)        |
+--------------------+-------------------+-----------------+
|  Class B           |  1st & 2nd Octets |  3rd, 4th       |
|                    | Example: 172.16.0.1|  (0.1)         |
+--------------------+-------------------+-----------------+
|  Class C           |  1st, 2nd & 3rd   |  4th Octet      |
|                    | Example: 192.168.1.1|  (1)          |
+--------------------+-------------------+-----------------+
```

Device Connection Scenario:

1. Device A IP Address: 17.0.0.1 (Class A)
2. Device B IP Address: 17.0.4.2 (Class A)

  Network ID for Class A: 17 (1st Octet)

  Since both devices have the same Network ID (17),
  they are in the same network and can connect directly.

Connection Summary:

```
+---------------------+-------------------------+---------------------+
|  Device A IP        |  Device B IP            |  Connection         |
+---------------------+-------------------------+---------------------+
|  17.0.0.1           |  17.0.4.2               |  Direct Connection  |
+---------------------+-------------------------+---------------------+
|  17.0.0.1           |  192.168.1.5            |  Requires Router    |
+---------------------+-------------------------+---------------------+
```

Explanation:
- If Device A and Device B are in the same network (same Network ID),
  they can connect directly.
- If they are on different networks (different Network IDs),
  a **Router** is required to connect them.

Router Usage:

```
+---------------------+-------------------------+---------------------+
|  Different Networks |  Example:               |  Connection         |
+---------------------+-------------------------+---------------------+
|  Device A Network ID |  17 (Class A)           |                     |
|  Device B Network ID |  192.168 (Class C)      |                     |
|  Router Needed      |  Yes                    |  Required to Connect |
+---------------------+-------------------------+---------------------+
```

**Subnetting:**

Divides a network into smaller and easier parts.

Example: A network with IP address 192.168.1.0/24 becomes subnets like 192.168.1.0/25 and 192.168.1.128/25.

***Example of Subnetting***

Given network: **192.168.1.0/24**

- **192.168.1.0/24** is a Class C network.
- **/24** indicates a subnet mask of **255.255.255.0**, meaning there are **8 bits for hosts** (32 total bits in IPv4–24 bits for the network portion = 8 bits for hosts).
- **192.168.1.0/24** provides **256 IP addresses** (from 192.168.1.0 to 192.168.1.255).

**Dividing into Smaller Subnets:**

To get two equal subnets, use /25 subnet masks. This gives 7 bits for hosts (32–25 = 7 bits for hosts).

1. **Subnet 1: 192.168.1.0/25**

- **Range**: 192.168.1.0 to 192.168.1.127
- **Subnet Mask**: 255.255.255.128
- **Total IPs**: 128 addresses (126 usable for hosts, as the first address is the network address and the last is the broadcast address).

**2. Subnet 2: 192.168.1.128/25**

- **Range**: 192.168.1.128 to 192.168.1.255
- **Subnet Mask**: 255.255.255.128
- **Total IPs**: 128 addresses (126 usable for hosts).

**Explanation**

Using a /25 mask instead of /24 divides the network into two parts. Each part has 128 IP addresses.

This creates smaller segments within the main network. Managing groups of hosts becomes easier.

**CIDR (Classless Inter-Domain Routing)**

CIDR (Classless Inter-Domain Routing) allocates IP addresses and routes IP traffic. This method replaces the old classful network system. CIDR helps use IP addresses more efficiently. It also makes routing simpler.

The table below shows the most common address and netmask combinations. It also provides important details about them.

| Prefix | Netmask | Number of Addresses |
|---|---|---|
| /32 | 255.255.255.255 | 1 |
| /25 | 255.255.255.128 | 128 |
| /24 | 255.255.255.0 | 256 |
| /23 | 255.255.254.0 | 512 |
| /16 | 255.255.0.0 | 65,536 |
| /15 | 255.254.0.0 | 131,072 |
| /8 | 255.0.0.0 | 16,777,216 |
| /0 | 0.0.0.0 | 4,294,967,296 |